Roman Povolotskii

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Education

Higher School of Economics St. Petersburg, Russia

Bachelor of Applied Mathematics and Computer Science *Class of 2026*

Relevant courses: Parallel Computing, Systems Design, Distributed Systems, Computer Architecture, Operating Systems, Databases, GPGPU, Software Engineering, Compilers

Skills

Programming Languages: Go, C/C++, Python, Rust
Production tools & Technologies: K8S, Docker, Kafka, PostgreSQL, Redis, NVIDIA Triton, Grafana, OTLP, Envoy
Skills: Concurrent Programming, Distributed Systems, Search Systems and Indexing, Performance Optimization

Experience

T-Bank Intern - Middle | *Golang, Rust, Python, Kafka, PostgreSQL, Redis, Triton* *Mar. 2025 – Now*

- **Team: Biometrics Platform Infrastructure**
- Benchmarked ANN search solutions and integrated Redis Search into production embedding retrieval services, achieving a **10x latency improvement** for over **60M biometric embeddings**
- Optimized production inference pipelines running on NVIDIA Triton Inference Server, increasing **inference throughput by 6.5x**
- Implemented backend services for **remote verification** and **remote identification** used in biometric workflows, **integrating with 3+ internal teams**
- Designed and implemented a production backend **service for deepfake detection**, integrating ML inference into a scalable microservice architecture
- Developed a **sidecar for encryption and token signing**, standardizing cryptographic operations across services
- Configured production infrastructure for **more than 10 services**, including Envoy, Kubernetes deployments, logging, metrics, and distributed tracing for backend services

Yandex Intern | *C++, Golang, PostgreSQL* *Jul. 2024 – Oct. 2024*

- **Team: authorization system infrastructure**
- Worked on backend services in a high-load authorization infrastructure handling **over 1 million RPS**
- Fixed bugs and improved stability of a **core authorization service** operating under production load
- Implemented functionality for **determining the master account** and integrated it into existing flows
- Extended a **secrets storage service** by adding new functionality and improving reliability of secret management

Projects

Diploma: Tokenizers for Search Systems — SereneDB | *C++* *Jan. 2026 – Now*

- **SereneDB** — Real-time Analytical Database
- **Researched and analyzed tokenization systems** used in modern search engines (Lucene, Tantivy, Elasticsearch), identifying trade-offs in performance, search quality, and multilingual support
- **Designed and optimized high-performance** text processing algorithms using **SIMD optimizations, compile-time specialization, reduced synchronization overhead and multithreading**
- Built benchmarking framework to evaluate throughput and latency across SereneDB, Apache Lucene and Tantivy
- **Achieved up to 4x higher throughput than Apache Lucene and 2.5x higher throughput than Tantivy**

Teaching

Teaching assistant on Algorithms and Data Structures (HSE) *Jan. 2025 – Jul. 2026*

- Conducted practical sessions for 1st-year Applied Mathematics and Computer Science students at HSE SPB
- **Reviewed and graded homework assignments,** provided feedback on solutions

Lecturer of course on Algorithms and Data Structures (competitive programming) *Jul. 2022, Jul. 2023*

- Developed the curriculum of course and conducted lectures and practices on algorithms and data structures course for **competitive programming students**. Created programming contests to qualify students for competitions

Other Achivements

1/2 ICPC (Northern Eurasia Finals) | *Third degree diploma* *Dec. 2024*

1/4 ICPC *Nov. 2022, Nov. 2023*